

# AssessClear Diagnostic Intelligence: Pre-Pilot Evaluation & Strategic Action Plan

Comprehensive Field Findings, Usability Assessment, Gap Analysis, Root Causes & Prioritized Product Roadmap

|                              |                                           |                                  |                                |
|------------------------------|-------------------------------------------|----------------------------------|--------------------------------|
| EVALUATION LEAD              | PRE-PILOT SCOPE                           | EVALUATED SAMPLE                 | MODEL ENGINE                   |
| Senior Evaluation Specialist | 9 Schools (Katni, Indore, Bhopal, Raisen) | ~351 Student Copies · Grades 6–8 | Qwen 3 (14B GPU) + Vision Core |

## 1. Executive Summary & Readiness Verdict

**Pre-Pilot Readiness Verdict: REQUIRES ATTENTION BEFORE PILOT (Conditional Green Light)**

The core diagnostic value proposition of automated multi-tiered feedback (1.0, 0.75, 0.40, 0.0) is proven and validated in field classrooms. However, critical software/vision defects (Q1/Q2 question skipping, digit misreading, rigid semantic matching) and operational bottlenecks ("double effort" workflow taking 80–90 min per 50 sheets) must be addressed before wide deployment.

**Validated Value Propositions**

- Constructive Diagnostic Feedback:**  
Hand-holds students with 3-part remedial steps (*What you did well, Think about this, Next steps*).
- Partial Attempt Credit:**  
Rewards partial understanding and correct mathematical approach (0.40 / 0.75 weights).
- Hindi OCR Relevance:**  
Directly supports Hindi-medium government classrooms across MP.

**Key Pilot Vulnerabilities**

- Question Dropping:**  
Vision pre-processing clips top/bottom questions (10 attempted → 7 evaluated).
- Teacher Overhead:**  
1–3 min review per paper creates a high administrative burden for classes of 30–50+ students.
- Zero-Network Failure:**  
Rural schools (MS Kharbai) freeze completely without offline sync.

**Table 1: Pre-Pilot Findings & District Evidence Matrix**

| District & Schools                                                                                                                | Sample Evaluated                                              | Key Field Strengths                                                                                                                                                                                           | Critical Observed Defects                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Indore</b><br>(Nandbag & Subhash Nagar)                                                                                        | 100 Students<br>(60 Math, 40 Hindi)<br>DIET<br>Observer       | <ul style="list-style-type: none"> <li>• Rapid batch upload efficient (<math>\leq 10</math> sheets).</li> <li>• 20–30s AI processing time.</li> <li>• Constructive feedback validated by DIET.</li> </ul>     | <ul style="list-style-type: none"> <li>• Q1 and Q2 skipped on first page.</li> <li>• Rigid logic rejected "3 months" when key was "90 days".</li> <li>• Incorrect arithmetic box sum marked correct.</li> <li>• Dashboard omitted Hindi and Grade 7–8 data.</li> </ul>                                                                                                                               |
| <b>Bhopal &amp; Raisen</b><br>(5 Schools: MS Green Park, MS Putlighar, HS Kanya in Bhopal; MS Kharbai & MS Bishankheda in Raisen) | ~160 Students<br>Grades 6–8<br>(Math & Hindi)                 | <ul style="list-style-type: none"> <li>• In-app direct scanning saves local phone memory.</li> <li>• 3-part structured feedback easy to understand.</li> <li>• Hindi language relevance confirmed.</li> </ul> | <ul style="list-style-type: none"> <li>• Zero network in MS Kharbai halted scanning (0s latency), while MS Bishankheda &amp; Bhopal schools processed normally (~26s AI processing).</li> <li>• Reviewing 30+ students individual feedback took &gt;80 min.</li> <li>• Downloading individual PDFs fills phone storage.</li> <li>• Feedback focused on "what" rather than "how to do it".</li> </ul> |
| <b>Katni</b><br>(EPES Purwar & MS Devri Hatai)                                                                                    | 91 Students<br>(32 Purwar, 59 Hatai)<br>Grade 6<br>Math/Hindi | <ul style="list-style-type: none"> <li>• Step-by-step remedial handholding praised.</li> <li>• Encourages learning beyond binary right/wrong.</li> </ul>                                                      | <ul style="list-style-type: none"> <li>• Dashboard showed 29 evaluations for 3 copies.</li> <li>• Session codes hard to retrieve without manual notes.</li> <li>• Digit misrecognition ( 9996 <math>\rightarrow</math> 3226 ).</li> <li>• Corrected pencil answers ignored; 200-eval cap drops data.</li> </ul>                                                                                      |

**Table 2: User Experience (UX) & Usability Evaluation Matrix**

| UX Dimension                       | Positive Design Elements                                                | Observed Field Friction & Usability Gaps                                                         |
|------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Navigation &amp; Access</b>     | Clean, intuitive role-based views for DLs, School Admins, and Teachers. | Dashboard filtering clutter; missing subject tabs; inconsistent grade views.                     |
| <b>Session Management</b>          | Amber session codes allow resuming work across different sittings.      | Codes are not saved in user profile; if not jotted down manually, sessions become orphaned.      |
| <b>Capture &amp; Ingestion</b>     | Rapid batch upload simplifies scanning up to 10 sheets simultaneously.  | One-by-one mode has page omission issues; lack of offline queue freezes app in low network.      |
| <b>Review &amp; Action</b>         | Rich misconception tags provide deep diagnostic insight.                | "Double Effort" dilemma: Teachers spend 3–4 min/paper transcribing remarks onto physical sheets. |
| <b>Language &amp; Localization</b> | Direct ingestion of handwritten Hindi scripts.                          | Hindi PDF output contains awkward grammar and phrasing; student feedback missing in PDF.         |

**Table 3: Gap Analysis Matrix (Expected Design vs Observed Reality)**

| System Dimension           | Designed / Intended Specification                        | Actual Pre-Pilot Experience                                   | Districts Affected  |
|----------------------------|----------------------------------------------------------|---------------------------------------------------------------|---------------------|
| <b>Question Detection</b>  | AI reads and grades 100% of questions from sheet.        | Skipped top/bottom questions (Q1, Q2); only 7 of 10 scored.   | Indore, Katni       |
| <b>Semantic Grading</b>    | Flexible diagnostic evaluation against learning outcome. | Rigid exact matching: marks 3 months wrong vs 90 days .       | Indore              |
| <b>Offline Resilience</b>  | PWA handles rural 3G/4G connectivity smoothly.           | Zero offline support; complete failure in zero-network areas. | Raisen (MS Kharbai) |
| <b>Dashboard Retention</b> | Continuous cohort SLO trend tracking.                    | Counts questions rather than copies; caps at ~200 items.      | Katni, Indore       |
| <b>PDF Export</b>          | Dual Teacher Instructions + Student Feedback PDF.        | Exported PDF shows only teacher feedback; student view lost.  | Katni               |
| <b>Teacher Autonomy</b>    | Teacher can review and calibrate AI assessments.         | No in-app 1-click override or score correction button.        | Indore, Bhopal      |

**Table 4: Issue, Root-Cause & Severity Analysis Table**

| Identified Issue                         | Direct Evidence Base                      | Technical / Pedagogical Root Cause                                                             | Severity             |
|------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------|----------------------|
| <b>Question Segmentation Defect</b>      | Q1 & Q2 skipped in Indore & Katni         | Auto-crop bounding box margins clip top/bottom headers of worksheet image.                     | <b>P0 (CRITICAL)</b> |
| <b>Handwriting &amp; Digit Errors</b>    | 9996 → 3226 , marked D read as B          | Insufficient prompt calibration and vision pre-processing for pencil handwriting.              | <b>P0 (CRITICAL)</b> |
| <b>Evaluation Count &amp; Cap Defect</b> | 29 evals for 3 copies; history drops >200 | Firestore schema tracks question IDs instead of student submissions; missing query pagination. | <b>P0 (CRITICAL)</b> |
| <b>Rural Network Dependency</b>          | MS Kharbai scanning lockout (Raisen)      | Absence of Service Worker offline caching and IndexedDB background sync queue.                 | <b>P1 (HIGH)</b>     |
| <b>Session Code Friction</b>             | Evaluators unable to resume sessions      | Active sessions not persisted in user profile or visible in dashboard dropdown.                | <b>P1 (HIGH)</b>     |
| <b>High Review Time Overhead</b>         | 3–4 min/paper review (80–90 min/class)    | Absence of aggregated Class-Level Summary Heatmaps; forces 1-by-1 screen review.               | <b>P1 (HIGH)</b>     |

**Table 5: Pilot Readiness Assessment & Strategic Risk Register**

| Readiness Status | Key Subsystem Diagnostics                                                              | Mandatory Pre-Pilot Mitigation Strategies                                                 |
|------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| P0 BLOCKER       | <b>AI Question Segmentation:</b> High risk of truncating valid student test responses. | Adjust crop padding, expand vision bounding boxes, and enforce schema count verification. |
| P0 BLOCKER       | <b>Data Schema &amp; Retention:</b> 200-evaluation cap truncates historical reporting. | Refactor database documents to track student submissions; paginate dashboard history.     |
| P1 HIGH RISK     | <b>Teacher Adoption Risk:</b> High time burden leads to rejection during pilot.        | Deploy Class-Level Summary UI to reduce total review time per batch to <5 minutes.        |
| P1 HIGH RISK     | <b>Connectivity Risk:</b> Field teams locked out during rural visits.                  | Implement offline scanning with automatic background sync upon network availability.      |

**Table 6: Strategic Action Roadmap & Prioritization Table (P0, P1, P2, P3)**

| Pri. | Strategic Recommendation                          | Operational Rationale                                                                | Timeline           |
|------|---------------------------------------------------|--------------------------------------------------------------------------------------|--------------------|
| P0   | <b>Fix Vision Segmentation &amp; Blur Gate</b>    | Eliminates Q1/Q2 skipping and prevents AI hallucinations on blurry images.           | Immediate (Week 1) |
| P0   | <b>Refactor Dashboard Schema &amp; Pagination</b> | Corrects evaluation counts (copies vs questions) and lifts the 200-eval limitation.  | Immediate (Week 1) |
| P0   | <b>Implement Offline-First PWA Caching</b>        | Enables uninterrupted scanning and local storage in zero-network rural schools.      | Immediate (Week 2) |
| P1   | <b>Build Class-Level Summary Dashboard</b>        | Reduces teacher review overhead from 80–90 min to <5 min for a class of 40 students. | Immediate (Week 2) |
| P1   | <b>Add Teacher Override / Edit Button</b>         | Empowers educators to correct AI scores, protecting trust and data integrity.        | Immediate (Week 2) |
| P1   | <b>Active Session Drawer in Profile</b>           | Eliminates lost session codes; allows 1-click session restoration.                   | Immediate (Week 2) |
| P2   | <b>Refine Hindi Prompts &amp; PDF Exports</b>     | Ensures high-quality Hindi grammar and formats dual Teacher/Student PDF exports.     | Pre-Pilot Polish   |
| P2   | <b>Deep Integration with Sakhi AI</b>             | Triggers automated WhatsApp lesson plans aligned to red-band class competencies.     | During Pilot       |

Table 7: Pilot Monitoring, Metrics & Evaluation Framework

| Evaluation Category | Key Performance Metric                                  | Pilot Target Benchmark                               | Verification Method                                              |
|---------------------|---------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------|
| Diagnostic Accuracy | Question Detection Rate<br>AI Scoring Agreement Rate    | > 99.0% detection<br>> 95.0% scoring agreement       | Compare AI scores against blind expert teacher evaluations.      |
| Operational Speed   | AI Processing Latency<br>Teacher Review Overhead        | < 25 sec per sheet<br>< 15 sec per student           | In-app telemetry logs and field time-motion observations.        |
| Field Reliability   | Offline Queue Sync Fidelity<br>Session Restoration Rate | 100% data sync<br>> 98.0% session recovery           | Track background sync logs across low-connectivity clusters.     |
| Teacher Engagement  | Weekly Active Usage<br>Sakhi AI Remedial Follow-up      | > 75% weekly adoption<br>> 60% remedial resource use | Dashboard analytics and Sakhi AI WhatsApp interaction telemetry. |

8. Final Analytical Conclusion

AssessClear represents a **transformative diagnostic innovation** for government school education, replacing delayed, high-overhead manual grading with real-time, multi-tiered misconception analysis. The pre-pilot exercise in Madhya Pradesh successfully proved the viability of mobile multimodal grading at ₹0.12/eval.

By executing the **PO blocker resolutions** (crop boundary correction, image quality gate, offline sync, and class-level summary screens) before full rollout, project leadership can ensure a high-fidelity, teacher-friendly, and statistically robust pilot deployment across the state.